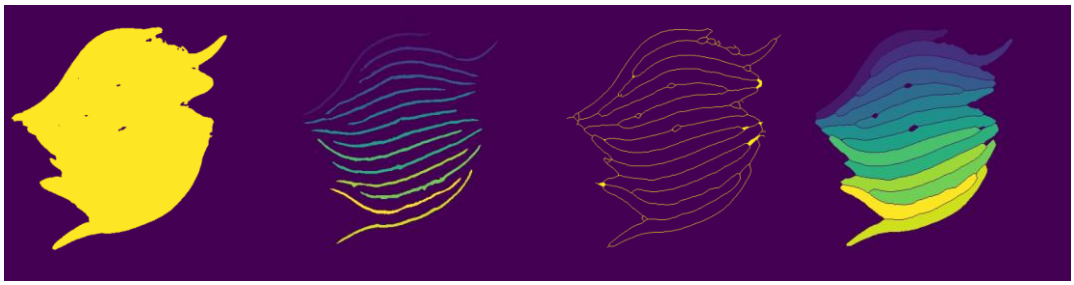
Worm Segmentor



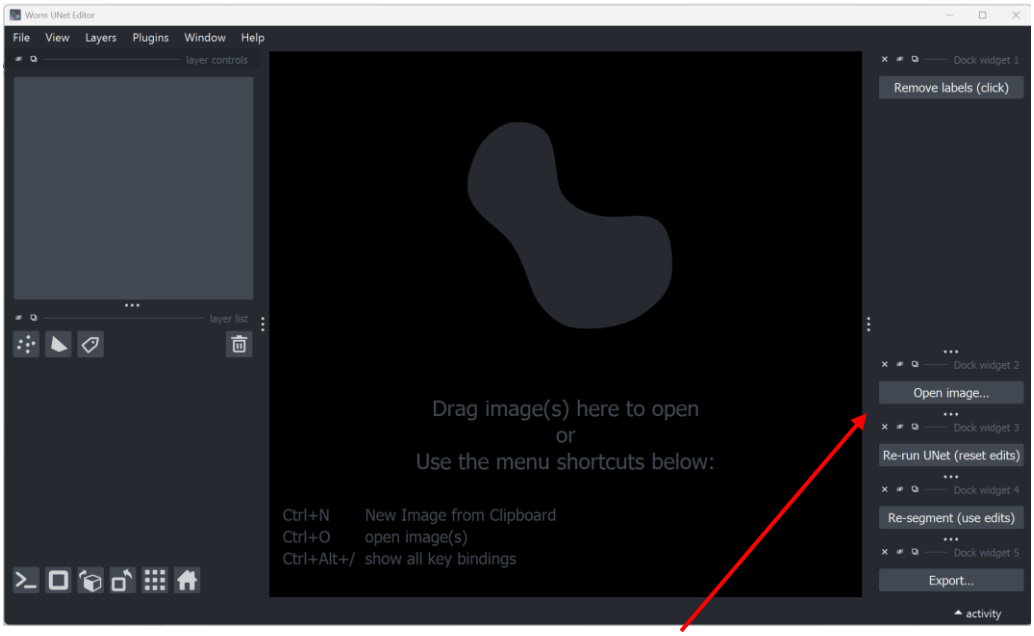
The program comes with 2 files. Keep them in the same folder.

Name	Date modified	Type	Size
unet_out_bf32_depth5_ep100.pth	9/26/2025 2:24 AM	PTH File	121,515 KB
WormGUIApp	11/11/2025 5:44 PM	Application	321,965 KB

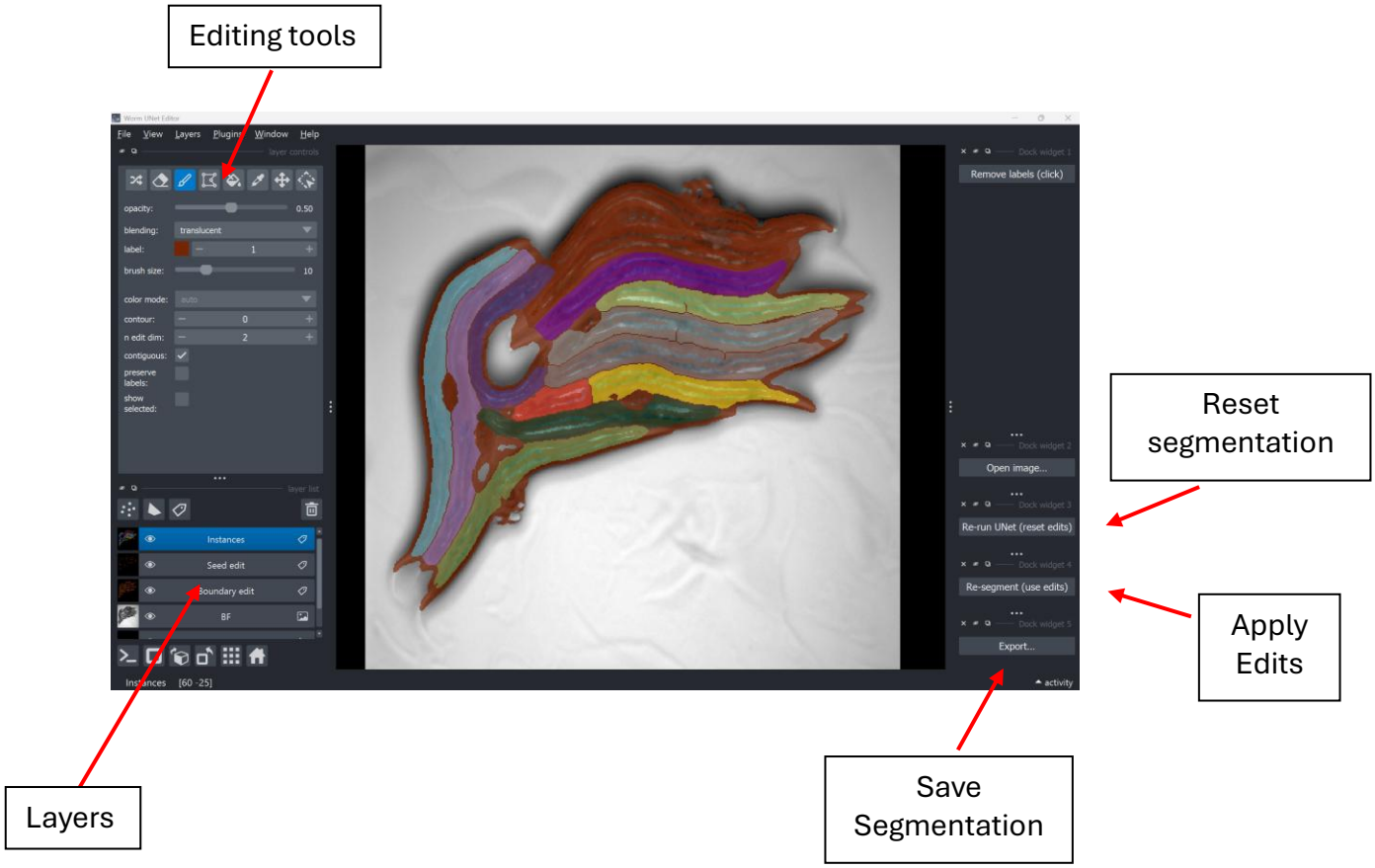
- WormGUIApp – the program executable
- unet_out_bf32_depth5_ep100.pth – the weights of the network.

To use WormSegmentor:

1. Run WormGUIApp. This window should open

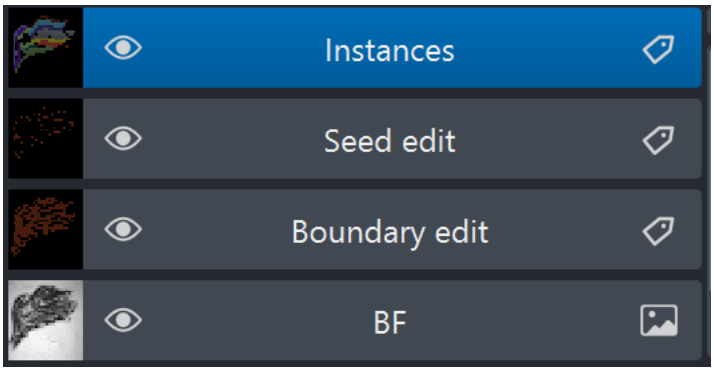


2. Click Open Image on the right side of the window and choose an image



3. Edit the image:

   The image consists of 4 layers.



   Instances: Colorful patches where each patch is a segmented worm

   Seed: Midlines of the worms in brown. Each worm must have at least one midline, but multiple are possible.

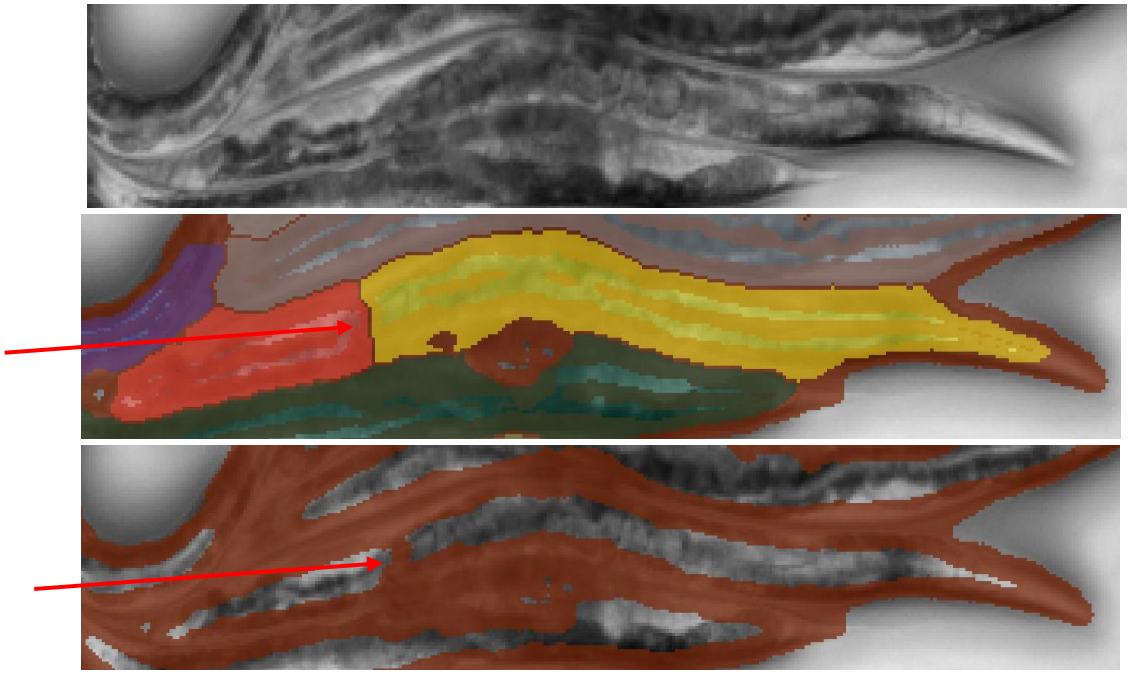
   Boundary: The edges of the worms in brown. There must be a complete closed line between 2 worms for them to be separated. There must **not** be a closed boundary inside a worm dividing it in half.

   BF: The brightfield image from the microscope.

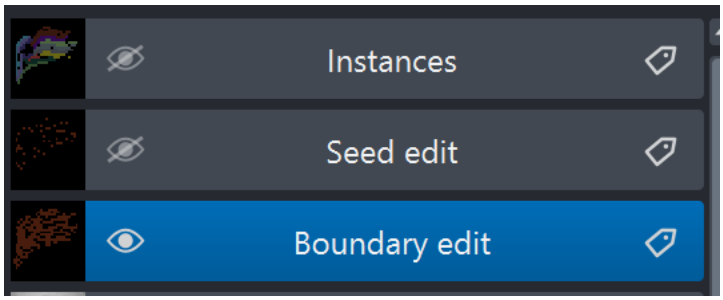
   Use the eye icon to show/hide a layer.

To edit a specific layer, the layer must be highlighted

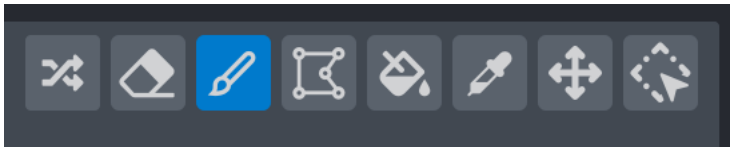
To fix an over-segmented worm remove the boundary in the middle of it



- Highlight the boundary layer

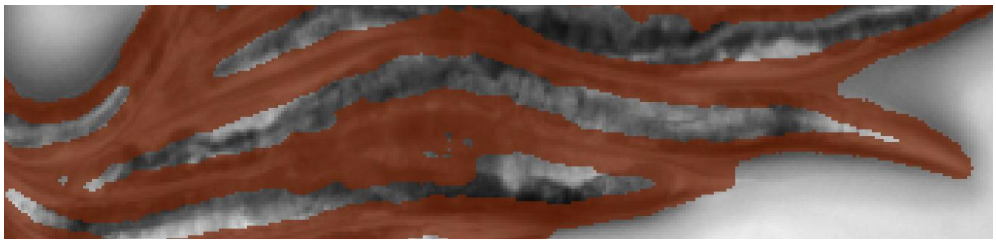


- Choose the brush tool

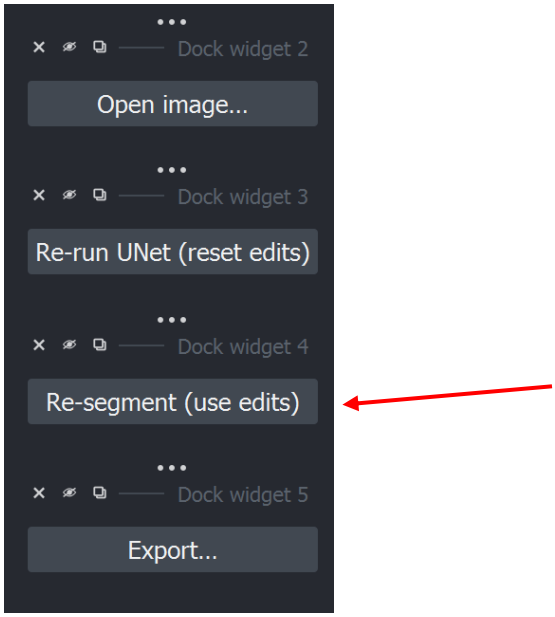


- To delete a line choose label = 0. You can also change the brush size

- Delete the extra line

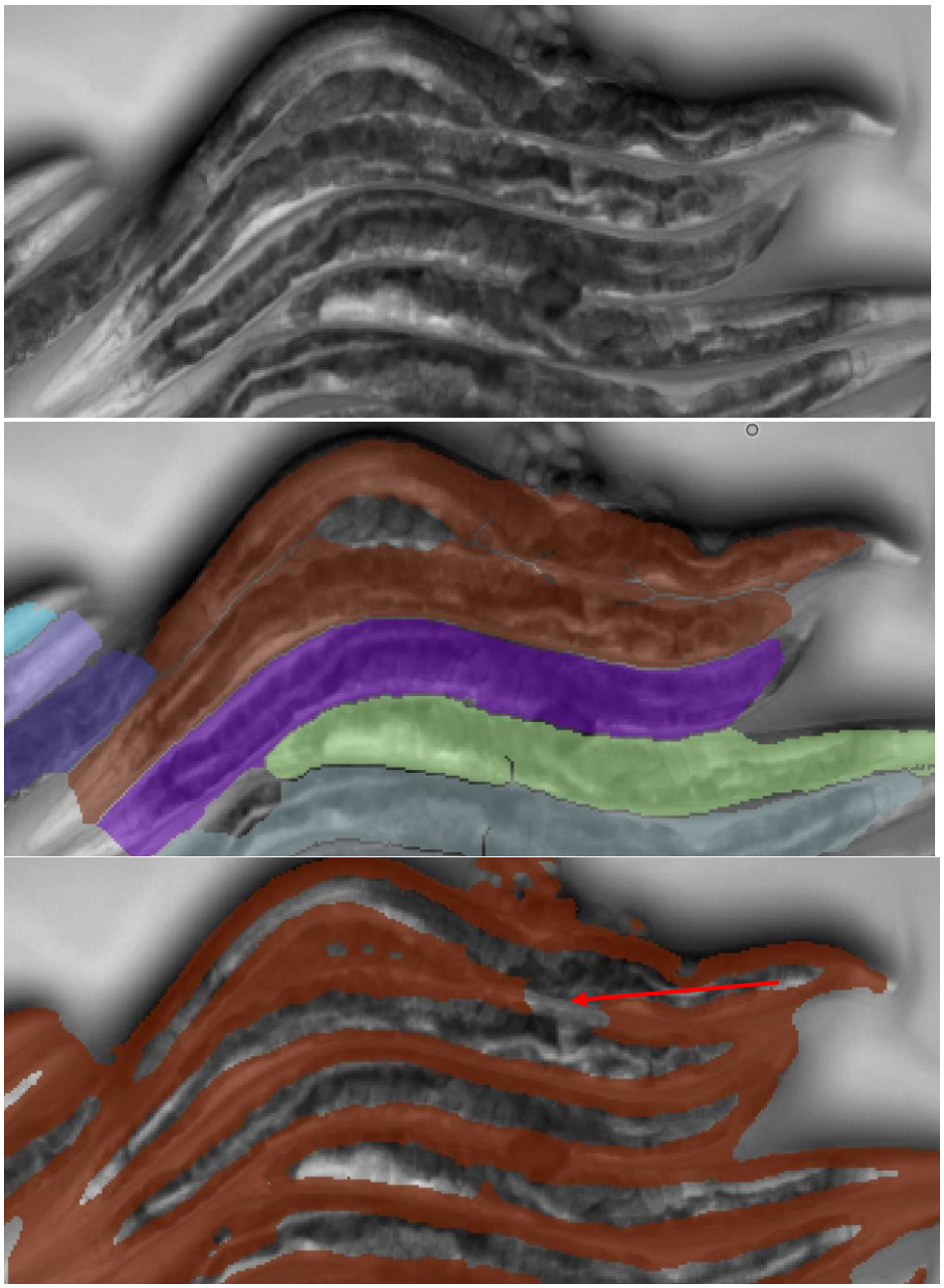


- Click Re-segment



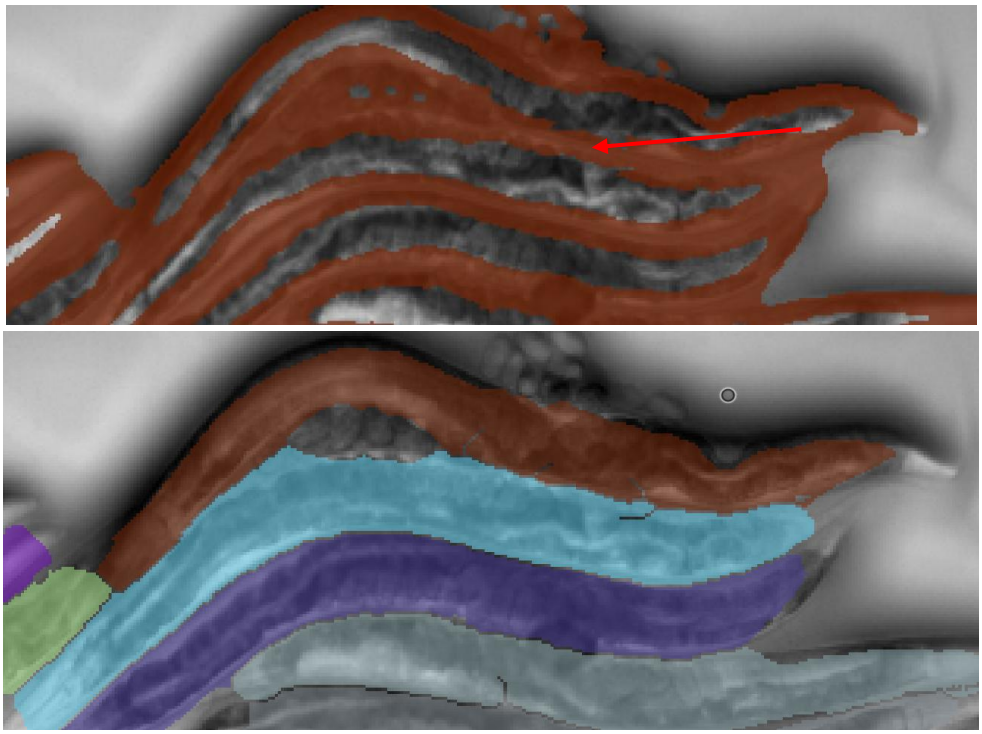
Do not click “Re-run Unet”, as it will reset all edits

To fix under-segmented worms complete the boundary between and around them



- Highlight the boundary layer and choose the brush tool
- Choose label = 1

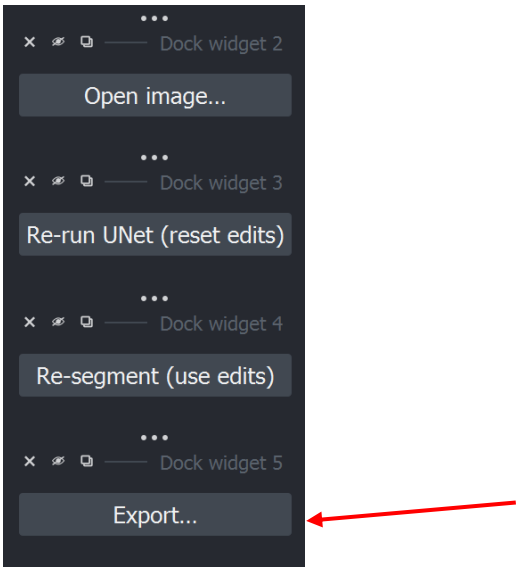
- Complete the boundary and click Re-segment



If the seeds of both worms are connected, disconnect them by highlighting the seed layer, changing the label to 0 and erasing the connected part

4. If there are stubborn small holes that refuse to be filled lower the value of the threshold slider and resegment

5. When done editing click “Export” and save



6. The output is 4 files

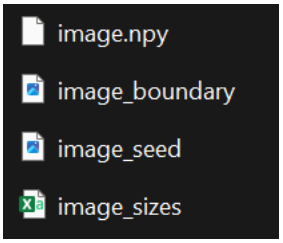


Image.npy: An image of the worm masks in a numpy array. Each worm is a block of its own label 1, 2, 3, etc..

Image_boundary: PNG image of the boundaries

Image_seed: PNG image of the midlines

Image_sized: CSV listing the measurements per worm (Ratio is length/width)

Area	Lengths	Widths	Ratios
5454	271.226	64.58194	4.199719
6574	279.663	70.9896	3.939492
5663	341.221	55.42424	6.15653

The files will be saved in the same folder as the image.